

CALHOUN COUNTY AP, TX, USA

WMO: 722209

Lat: 28.654N

Lon: 96.681W

Elev: 9

StdP: 101.22

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 12986

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.4	2.3	-7.0	2.1	7.1	-4.4	2.6	8.1	11.9	15.9	11.1	15.6	4.1	0	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.3	35.7	25.0	34.7	25.3	33.7	25.4	27.5	31.8	27.2	31.4	26.8	30.9	4.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	21.9	29.1	26.2	21.6	29.0	25.8	21.1	28.8	87.1	32.0	85.3	31.3	83.5	31.0	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.3	8.4	DB	-2.3	37.5	2.9	1.3	-4.5	38.4	-6.2	39.2	-7.8	39.9	-9.9	40.8	(4)
(5)			WB	-3.7	27.7	2.6	0.9	-5.6	28.4	-7.1	28.9	-8.6	29.3	-10.5	30.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.8	12.7	15.1	18.3	21.6	25.1	27.9	28.8	29.3	27.3	23.3	17.6	14.0	(6)	
	DBStd	6.89	5.01	5.12	4.38	3.32	2.70	1.51	1.39	1.54	2.18	3.99	4.84	5.20	(7)	
	HDD10.0	74	29	14	3	1	0	0	0	0	0	0	6	21	(8)	
	HDD18.3	593	183	111	56	11	1	0	0	0	0	9	69	152	(9)	
	CDD10.0	4383	114	159	261	348	468	538	583	599	519	412	235	146	(10)	
	CDD18.3	1859	9	22	55	109	211	288	325	341	269	163	49	18	(11)	
	CDH23.3	18607	33	70	201	635	1863	3203	3811	4136	2813	1483	286	74	(12)	
(13) Wind	CDH26.7	7557	1	6	18	109	575	1385	1753	2011	1177	486	36	2	(13)	
	WSAvg	3.9	4.2	4.5	4.7	4.8	4.4	3.4	3.1	3.1	3.2	3.7	3.9	4.0	(14)	
	PrecAvg	1082	80	56	78	64	107	103	103	72	137	118	89	67	(15)	
	PrecMax	1617	256	180	311	173	326	312	434	158	333	311	227	199	(16)	
	PrecMin	583	2	11	6	3	0	1	5	9	39	19	2	15	(17)	
	PrecStd	302	61	45	68	49	77	76	105	49	80	79	66	48	(18)	
	DB	26.1	27.3	28.4	32.0	32.9	36.0	36.3	37.0	36.3	33.4	29.1	26.9		(19)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	19.5	20.6	20.9	19.2	24.4	25.0	24.9	24.5	24.6	25.3	22.5	20.9		(20)	
	2% DB	24.0	25.3	26.9	29.0	32.2	34.0	35.1	35.9	34.1	32.2	27.7	25.3		(21)	
	MCWB	19.4	20.1	21.2	21.5	24.4	25.2	25.2	25.1	25.1	24.9	22.0	20.5		(22)	
	5% DB	22.6	23.9	25.7	27.7	31.0	32.9	33.9	34.9	32.9	31.0	26.7	23.8		(23)	
	MCWB	18.5	19.9	20.5	22.0	24.1	25.2	25.5	25.5	24.9	24.2	21.7	20.5		(24)	
	10% DB	21.1	22.5	24.1	26.8	29.9	32.3	33.0	33.9	32.2	29.5	25.1	22.4		(25)	
	MCWB	17.9	19.6	20.2	21.5	24.0	25.2	25.5	25.6	24.9	23.4	20.9	19.8		(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	22.0	22.9	23.4	25.2	26.9	28.0	27.9	28.0	27.9	26.9	25.5	23.3		(27)	
	MCDB	23.8	25.3	26.5	27.6	30.2	31.7	32.4	32.7	32.1	30.8	27.9	25.3		(28)	
	2% WB	21.0	21.8	22.7	24.4	26.2	27.3	27.3	27.4	27.3	26.3	24.0	22.5		(29)	
	MCDB	22.8	23.7	25.2	27.1	29.2	31.0	31.8	32.2	31.1	30.0	26.0	24.2		(30)	
	5% WB	20.0	21.1	22.0	23.6	25.6	26.6	26.8	27.0	26.8	25.7	22.9	21.5		(31)	
	MCDB	22.0	22.9	24.1	26.2	28.8	30.4	31.1	31.6	30.5	29.1	25.3	23.0		(32)	
	10% WB	18.7	20.2	21.3	23.0	25.1	26.0	26.5	26.6	26.2	25.0	21.8	20.3		(33)	
(34) Mean Daily Temperature Range	MCDB	20.4	21.9	23.3	25.4	28.3	29.9	30.6	31.0	29.9	28.3	24.0	21.9		(34)	
	MDBR	10.8	9.9	9.4	9.0	8.5	8.5	8.7	9.3	9.1	10.5	10.6	10.3		(35)	
	5% DB	12.2	10.3	9.1	9.5	9.4	9.3	9.8	10.3	10.0	9.4	10.5	11.2		(36)	
	MCWBR	8.1	6.4	4.8	4.8	3.5	2.8	2.8	2.9	3.4	4.1	6.1	7.6		(37)	
	5% WB	10.1	8.7	7.9	7.4	7.1	8.0	8.5	8.8	8.2	8.5	8.8	9.4		(38)	
	MCWBR	7.8	6.4	4.8	4.2	3.3	3.2	3.1	3.1	3.2	4.5	6.2	7.6		(39)	
	taub	0.352	0.368	0.384	0.441	0.483	0.459	0.483	0.457	0.445	0.391	0.380	0.343		(40)	
(41) Clear-Sky Solar Irradiance	taud	2.481	2.441	2.391	2.240	2.153	2.285	2.229	2.308	2.304	2.447	2.422	2.523		(41)	
	Ebn at Noon	886	901	905	860	820	835	814	834	834	865	846	881		(42)	
	Edh at Noon	94	105	117	141	154	134	142	130	127	104	99	86		(43)	
	RadAvg	2.94	3.42	4.53	5.39	6.08	6.64	6.54	6.22	5.20	4.49	3.38	2.67		(44)	
(45) All-Sky Solar Radiation	RadStd	0.32	0.56	0.41	0.48	0.41	0.40	0.59	0.35	0.47	0.47	0.37	0.33		(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	+93

Nomenclature: See separate page